Financial Econometrics:
Review on OLS and Linear Regression Models

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Note: This lecture note provides the summary of key topics covered in an intro-level Econometrics course. It covers Ch 2,3,4,8,15 of Introductory Econometrics: A Modern Approach (6th edition) by Jeffrey M. Woodldridge.

Outline

Motivation

- When is the Ordinary Least Squares estimator (OLS) reliable for the estimation of a linear regression model?
- If OLS is problematic, can we employ other estimation methods?
- The other methods are Generalized Least Squares estimator (GLS) and 2 Stage Least Squares estimator (2SLS).

Outline

OLS - As minimizing residuals

We consider a simple linear regression model to review OLS.

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

- Suppose that we have n observations for two variables: $(x_1, \ldots x_n)$ and $(y_1, \ldots, y_n)$
- OLS is the method to find the line $\left(\widehat{\beta}_0 + \widehat{\beta}_1 x\right)$ that best fits the data. The estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are the OLS estimates.
- $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ is the fitted value for y_i. Also, $\widehat{y}_i$ can be interpreted as the the value of y_i predicted by the model.
- The residual is $\widehat{u}_i = y_i - \widehat{y}_i$. Also, $\widehat{u}_i$ can be interpreted as the part of y_i that cannot be predicted or explained by the model.

OLS - As minimizing residuals

- OLS: Find the values of β_0 and β_1 such that the overall magnitude of the residuals is minimized.
- We don't care if the residual $\widehat{u}_i$ is positive or negative, we want it to be small.
- So we square it: $\widehat{u}_i^2$
- Why not using the absolute value? Good statistical reasons + harder to work with $|\cdot|$
- We have n residuals. So our goal is to minimize $\sum_{i=1}^{n} \widehat{u}_i^2$ (Sum of Squared Residuals, SSR).

OLS - As minimizing residuals

Objective function for the OLS estimator:

$$\min_{\widehat{\beta}_0,\widehat{\beta}_1} \sum_{i=1}^{n} \widehat{u}_i^2 = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 + \widehat{\beta}_1 x_i\right)^2$$

Using some basic calculus (getting the first derivatives with respect to $\widehat{\beta}_0, \widehat{\beta}_1$ and equating the two first derivatives to zero), we can obtain the analytical expression for the OLS estimator:

$$\widehat{\beta}_1^* = \frac{\sum_{i=1}^{n} (x_i - \overline{x_i})(y_i - \overline{y_i})}{\sum_{i=1}^{n} (x_i - \overline{x_i})^2}$$

$$= \frac{\frac{1}{n}\sum_{i=1}^{n} (x_i - \overline{x_i})(y_i - \overline{y_i})}{\frac{1}{n}\sum_{i=1}^{n} (x_i - \overline{x_i})^2} = \frac{\text{Sample covariance } (\mathrm{x}, \mathrm{y})}{\text{Sample variance } (\mathrm{x})}$$

$$\widehat{\beta}_0^* = \overline{y_i} - \widehat{\beta}_1^* \overline{x_i}$$

OLS - As minimizing residuals

https://ryansafner.shinyapps.io/ols_estimation_by_min_sse/

Outline

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots \beta_k x_{k,i} + e_i$$

- Assumption 1: linearity in parameters
- Assumption 2: $E[e_i | x_{1,i}, x_{2,i}, ..., x_{k,i}] = 0 \;\; \forall i$
- Assumption 3: $E[e_i^2 | x_{1,i}, x_{2,i}, ..., x_{k,i}] = \sigma^2 \;\; \forall i$ (homoscedasticity)
- Assumption 4: $Cov[e_i e_j] = E[e_i e_j] = 0 \;\; \forall i \neq j$ (Random sampling)
- Assumption 5: No exact linear relation among x's. (no perfect multi-collinearity)
- Assumption 6: e_i follows a normal distribution.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots \beta_k x_{k,i} + e_i$$

- The set of the first 5 assumptions is referred to as the classical model assumption.
- Under the classical model assumptions, OLS is the Best Linear Unbiased Estimator (BLUE).
- This is the Gauss-Markov theorem.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots \beta_k x_{k,i} + e_i$$

- Assumption 6 allows us to make exact statistical inference.
- Without assumption 6, we can make approximate statistical inference based on asymptotic theories. This approximation is fine as long as the sample size n is large enough.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots \beta_k x_{k,i} + e_i$$

- If the assumption 2 is violated ($E[e_i|x_1, x_2, ..., x_k] \neq 0$), OLS no longer works. In this case, it becomes a biased estimator.
- Question is ... when is the assumption 2 violated?

Regression model and classical assumptions

(True) Population:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots + \beta_k x_{k,i} + \beta_{k+1} x_{k+1,i} + e_i^*$$

Used Model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots + \beta_k x_{k,i} + e_i$$

- Remember that the error term e_i in the model now contains $\beta_{k+1} x_{k+1,i} + e_i^*$.
- If the omitted variable $x_{k+1,i}$ is correlated with the included variable x_1, the OLS estimator of $\hat{\beta}_1$ is biased.
- The biased $\hat{\beta}_1$ captures $\beta_1 + \beta_{k+1} \frac{\Delta x_{k+1,i}}{\Delta x_{1,i}}$.
- Even if the covariance between $x_{k+1,i}$ and other x variables are not correlated, the variances of $\hat{\beta}_2, \hat{\beta}_3, \ldots, \hat{\beta}_k$ will be inflated because of the omitted variable $x_{k+1,i}$.

Regression model and classical assumptions

Statically,

$$Cov[x_{1,i}, x_{k+1,i}] \neq 0$$

$$Cov[x_{1,i}, e_i] = Cov[x_{1,i}, \beta_{k+1} x_{k+1,i} + e_i^*] \neq 0$$

For simplicity, assume that $E[x_{1,i}] = 0$. Then,

$$Cov[x_{1,i}, e_i] = E[x_{1,i} e_i] = E[E[x_{1,i} e_i | x_{1,i}]] = E[x_{1,i} E[e_i | x_{1,i}]]$$

So, if $Cov[x_{1,i}, e_i] \neq 0$, then $E[e_i | x_{1,i}] \neq 0$.

Regression model and classical assumptions

(True) Population:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots + \beta_k x_{k,i} + \beta_{k+1} x_{k+1,i} + e_i^*$$

Used Model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots + \beta_k x_{k,i} + e_i$$

- What do we do when there exists the omitted variable bias?
- Find x_{k+1} (collect data) and put it in the model!
- If no data is available, use an alternative (or alternatives)! That is an instrument variable!
- Use the 2SLS approach for correct estimation.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots \beta_k x_{k,i} + e_i$$

- If assumption 5 (no perfect multicollinearity) is violated, OLS does not work.
- If assumption 5 is not exactly violated but multicollinearity is very strong, the variance of the OLS estimator is inflated. The common symptom is a spuriously low t value.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots \beta_k x_{k,i} + e_i$$

- If assumption 3 or 4 are violated...
- That is, there exists heteroscedasticity or non-zero correlations among errors, or both!
- In the case, OLS is no longer BLUE.
- The Generalized Least Squares (GLS) estimator is better in the sense that the estimation uncertainty is smaller than the OLS.

Regression model and classical assumptions

A general multiple regression model is given by

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots \beta_k x_{k,i} + e_i$$

- In order to use GLS, we should know the structure of heteroscedasticity or the structure of the correlations among errors.
- Feasible GLS is the GLS that use the estimated structure of heteroscedasticity or the estimated structure of the correlations among errors.

Outline

Frisch-Waugh-Lovell (FWL) theorem

Consider the following multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i} + \cdots + e_i$$

- OLS estimator with covariates: $\hat{\beta}_1, \hat{\beta}_2, ..., \hat{\beta}_k$
- There is another way to get the estimates of beta's.

Frisch-Waugh-Lovell (FWL) theorem

Consider the following multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i} + \cdots + e_i$$

- Firstly, x_{1i} is regressed on all the remaining independent variables

$$x_{1i} = \gamma_{1,0} + \gamma_{1,2} x_{2,i} + \gamma_{1,3} x_{3,i} + \cdots + \gamma_{1,k} x_{ki} + u_{1,i}$$

  And then, we get the OLS residual $\hat{u}_{1,i}$.
- Secondly, $x_{2,i}$ is regressed on all the remaining independent variables

$$x_{2,i} = \gamma_{2,0} + \gamma_{2,1} x_{1,i} + \gamma_{2,3} x_{3,i} + \cdots + \gamma_{2,k} x_{ki} + u_{2,i}$$

  And then, we get the OLS residual $\hat{u}_{2,i}$.
- Repeat this process for all other x's to get residuals.

Frisch-Waugh-Lovell (FWL) theorem

Consider the new multiple regression model with the obtained residuals:

$$y_i = \beta_0 + \beta_1 \hat{u}_{1,i} + \cdots + \hat{\beta}_k \hat{u}_{k,i} + \cdots + e_i$$

- OLS estimator with the residuals: $\hat{\beta}_1^*, \hat{\beta}_2^*, ..., \hat{\beta}_k^*$
- The FWL theorem tells us that $\hat{\beta}_1 = \hat{\beta}_1^*, \hat{\beta}_2 = \hat{\beta}_2^*, ..., \hat{\beta}_k = \hat{\beta}_k^*$

Frisch-Waugh-Lovell (FWL) theorem

Graphically, we can express this...

Next, let's talk about return predictability using a linear regression model!